Classifying Severe Weather Events from NOAA Storm Report Features

CISC 484: Introduction to Machine Learning — Phase 1 Proposal

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1 Problem Definition and Scientific Hypothesis

Severe weather events such as tornadoes, hail storms, and flash floods cause billions of dollars in damage across the United States annually. The NOAA Storm Events Database is a detailed record of significant weather events reported by National Weather Service offices nationwide. This project investigates whether machine learning models can accurately classify severe weather event types from contextual event characteristics (location, timing, zone type), and whether an attention-based deep learning architecture provides a meaningful advantage over gradient-boosted trees on this structured tabular classification task.

Task: Multi-class classification of severe weather event types from tabular contextual features including geographic location, temporal attributes, and event duration.

Hypothesis: We hypothesize that a TabNet classifier, which uses sequential attention to perform instance-wise feature selection, will outperform an XGBoost baseline on this multi-class weather classification task, because the learned attention masks can capture non-linear feature interactions and context-dependent feature relevance that fixed tree splits cannot represent. We expect TabNet to show the largest gains on event types distinguished by subtle combinations of geographic and temporal features (e.g., separating Thunderstorm Wind from Hail, which co-occur in similar conditions).

2 Data Audit and Initial EDA

2.1 Dataset Overview

We use the NOAA Storm Events Database, specifically the bulk CSV detail files for 2023 and 2024 from the National Centers for Environmental Information. The combined dataset contains 145,394 event records across 54 distinct event types and 68 U.S. states and territories. Each record includes the event type, location (state, county, coordinates), timing, and damage estimates.

Property	Value
Source	NOAA Storm Events Database (Bulk CSV)
Total Instances	145,394
Features (Columns)	51 per record
Target Classes	54 event types (top 10 used for classification)
Time Range	2023–2024
Geographic Scope	68 U.S. states and territories

2.2 Scale Verification

The combined dataset contains 145,394 labeled instances, well above the 10,000 minimum requirement. For classification, we retain the top 10 event types, which cover 81.0% of all records (117,717 instances). The remaining 44 rare types are excluded to avoid extreme class sparsity.

2.3 EDA Visualizations

Figure 1 shows the class distribution across the top 12 event types. The data exhibits significant class imbalance: Thunderstorm Wind accounts for 28.3% of all records (41,177 out of 145,394), while Flood represents only 3.7%. This imbalance will be addressed through stratified sampling and class-weighted loss functions.

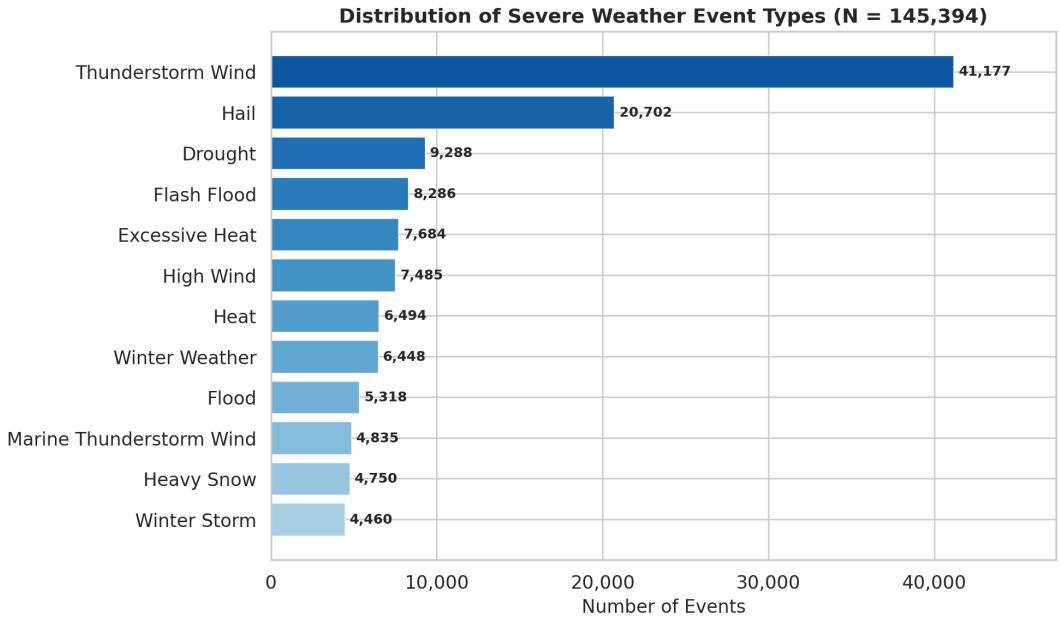

Figure 1: Class distribution of severe weather event types (2023–2024).

Figure 2 shows the monthly distribution of the five most frequent event types. Thunderstorm wind and hail concentrate in the May–August warm season, while drought persists into fall and excessive heat peaks in July–August. This strong seasonal structure confirms that month will be a discriminative feature.

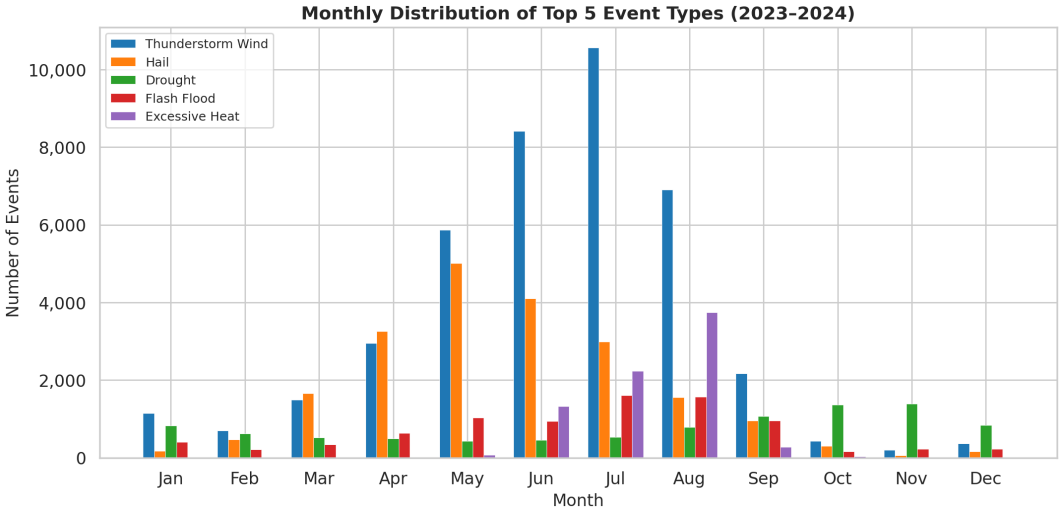

Figure 2: Monthly distribution of the top 5 event types (2023–2024).

2.4 Preprocessing Strategy

Our preprocessing pipeline has three stages. First, we engineer features from raw columns: event duration (in minutes from start/end timestamps), hour of day, and month. Latitude and longitude missing values (39.7% of records) will be imputed using per-state median coordinates. Second, categorical features (state, CZ_TYPE, WFO) will be label-encoded, which both XGBoost and TabNet handle natively. Third, continuous features will be standardized to zero mean and unit variance using training-set statistics only to prevent data leakage.

3 Comparative Methodology

3.1 Feature Set

To avoid data leakage, we exclude all outcome-derived columns (damage estimates, injuries, deaths, magnitude measurements, and tornado-specific fields like TOR_F_SCALE). Notably, the MAGNITUDE column is populated for 100% of wind and hail events but 0% of all other types, making it a near-perfect proxy for the target variable. Our features are restricted to contextual attributes known independently of the event classification:

- **Month:** month of event start (categorical, 12 levels).
- **Hour of Day:** hour of event start (integer, 0–23).
- **State:** U.S. state or territory (categorical, 68 levels).
- **Latitude / Longitude:** geographic coordinates of the event origin.
- **Duration:** elapsed time from event start to end (continuous, minutes).
- **CZ_TYPE:** county (C) or zone (Z) designation.
- **WFO:** issuing Weather Forecast Office (categorical, $\sim$120 levels).

3.2 Phase 1 Baseline: XGBoost

Our baseline is an XGBoost gradient-boosted tree classifier. Each instance is a flat feature vector of the attributes above. We will tune the learning rate, max depth, number of estimators, and subsample ratio using 5-fold stratified cross-validation on the training set.

3.3 Phase 2 Advanced Model: TabNet

Our advanced model is TabNet, an attention-based deep learning architecture for tabular data (Arik & Pfister, 2021). Unlike XGBoost's fixed decision splits, TabNet uses sequential attention for instance-wise feature selection at each decision step, learning to focus on different feature subsets depending on the input. It offers a scikit-learn-compatible API and built-in interpretability: the attention masks reveal which features drove each prediction, enabling direct comparison with XGBoost's feature importance. Key hyperparameters include the number of decision steps, attention embedding dimension, and learning rate.

4 Evaluation and Success Metrics

We evaluate both models using metrics suited to imbalanced multi-class classification:

- **Macro F1-Score:** Our primary metric. Macro-averaging gives equal weight to each class regardless of frequency, preventing the majority class from dominating the score.
- **Per-Class ROC-AUC:** One-vs-rest curves per event type, revealing which classes benefit most from TabNet's attention versus XGBoost's tree splits.
- **Confusion Matrix:** To identify systematic error patterns and whether TabNet reduces specific confusions (e.g., Thunderstorm Wind vs. Hail).

Our validation strategy is stratified 5-fold cross-validation for hyperparameter tuning, with a held-out 20% test set for final evaluation. We will also perform a temporal split (train on 2023, test on 2024) to confirm generalization across years.